

Sketching $y = x^2 + bx + c$

e.g. $y = x^2 - 4x - 12$

y -intercept: $x = 0, y = -12$

x -intercepts: $y = 0, x^2 - 4x - 12 = 0$

Factorise $(x - 6)(x + 2) = 0$

Null Factor Law

$x - 6 = 0$ or $x + 2 = 0$

$x = 6$ or $x = -2$

Turning point: halfway between x -intercepts

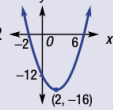
$$x = \frac{6 + (-2)}{2} = 2$$

Substitute $x = 2$:

$$y = 2^2 - 4(2) - 12$$

$$= -16$$

$(2, -16)$ minimum



Sketching $y = a(x - h)^2 + k$

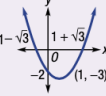
Any quadratic can be written in turning point form by completing the square.

e.g. sketch $y = (x - 1)^2 - 3$

Turning point $(1, -3)$ minimum

y -intercept: $x = 0, y = 1 - 3$

$$= -2$$



x -intercepts: $y = 0$

$$0 = (x - 1)^2 - 3$$

$$3 = (x - 1)^2$$

$$\pm\sqrt{3} = x - 1$$

$$\therefore x = 1 + \sqrt{3}, 1 - \sqrt{3}$$

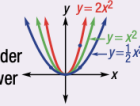
(alternatively use the Null Factor Law)

Transformations and turning point form

$y = ax^2$ dilates graph compared to $y = x^2$

$0 < a < 1$ graph is wider

$a > 1$ graph is narrower



$a < 0$ graph is inverted (reflected in x -axis)

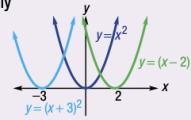


$y = (x - h)^2$ translates $y = x^2$

h units horizontally

$h > 0$ move right

$h < 0$ move left

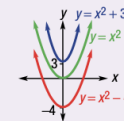


$y = x^2 + k$ translates $y = x^2$

k units vertically

$k > 0$ move up

$k < 0$ move down



Turning point form

$y = a(x - h)^2 + k$ has turning point at (h, k)

Quadratic formula

For $y = ax^2 + bx + c$, the quadratic formula can be used to find x -intercepts.

i.e. if $ax^2 + bx + c = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The discriminant $\Delta = b^2 - 4ac$ tells how many x -intercepts:

$\Delta > 0$

two x -intercepts



$\Delta = 0$

one x -intercept



$\Delta < 0$

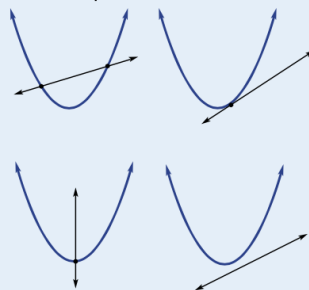
no x -intercepts



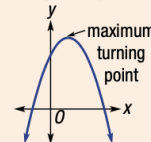
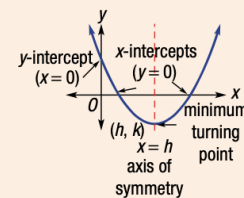
Parabolas, rates of change and variation

Lines and parabolas

Using substitution we can solve simultaneously to find 0, 1 or 2 intersection points.



Key features



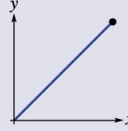
Quadratic applications

Form a quadratic model based on the formulation in the question. Sketch the graph for suitable values of x , to locate max/min and intercepts.

Direct variation and inverse variation

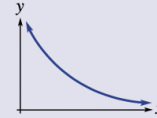
Direct variation: $y = kx$

- Both variables increase or decrease together.
- Graph is a straight line through (0, 0).



Inverse (indirect) variation: $y = \frac{k}{x}$ or $xy = k$

- If one variable increases, the other decreases.
- Graph is a hyperbola shape.

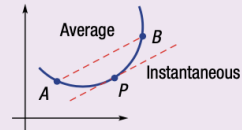


Rates of change

Average and instantaneous rates of change

The average rate of change between two points is equal to the gradient of the line joining the two points.

The instantaneous rate of change is equal to the gradient at a particular point and can be approximated by the average rate of change between two selected points.



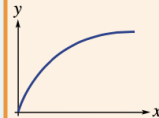
Rates of change

The shape of a graph shows how both y and the rate of change of y (i.e. the gradient) vary.

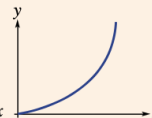
Straight lines have a constant rate of change; curves have a varying rate of change.

Description of graphs:

- y is increasing or decreasing or constant.
- Rate of change is positive or negative or zero.
- Rate of change is increasing or decreasing or constant.



Positive decreasing gradient



Positive increasing gradient

Chapter checklist with success criteria

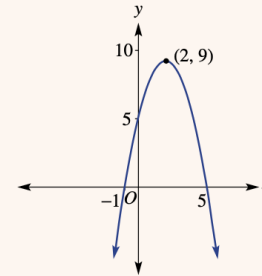
A printable version of this checklist is available in the Interactive Textbook



7A

1. I can identify the key features of a parabola.

e.g. For the graph shown determine the turning point (and whether it's a maximum or a minimum), the axis of symmetry, the x -intercepts and the y -intercept.

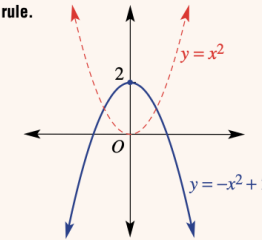


7A

2. I can find key features of transformed graphs from a rule.

e.g. Complete the table for the graph shown

Formula	Max or min	Turning point	y -value when $x = 1$
$y = -x^2 + 2$			



7B

3. I can sketch a quadratic relation involving a dilation or reflection.

e.g. Sketch the graph of $y = -4x^2$, labelling the turning point and one other point.



7B

4. I can sketch a quadratic relation involving translations.

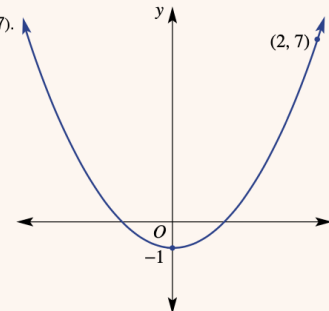
e.g. Sketch the graph of $y = (x - 1)^2 + 3$, labelling the turning point and y -intercept.

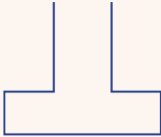


7B

5. I can find a rule in turning point form from a quadratic graph.

e.g. Find a rule for this parabola with turning point $(0, -1)$ and another point $(2, 7)$.

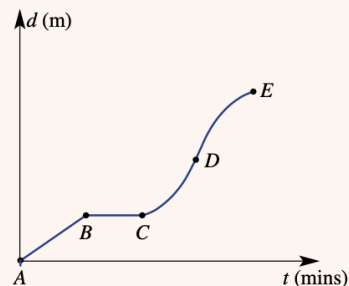


7C	6. I can use the x-intercepts to find the turning point and sketch a quadratic graph. e.g. Sketch the graph of $y = x^2 - 2x - 3$, labelling the intercepts and the turning point.	☐
7C	7. I can sketch a quadratic graph that is a perfect square. e.g. Sketch the graph of $y = x^2 - 4x + 4$, labelling the intercepts and turning point.	☐
7D	8. I can determine the key features of a graph in turning point form. e.g. For $y = 2(x - 1)^2 - 18$, determine the turning point, y -intercept and any x -intercepts.	☐
7D	9. I can sketch a quadratic graph by first completing the square. e.g. Sketch $y = x^2 + 8x + 20$ by first completing the square.	☐
7E	10. I can use the discriminant to find the number of x-intercepts of a quadratic graph. e.g. Determine the number of x -intercepts of the parabola given by $y = 2x^2 + 3x + 6$.	☐
7E	11. I can find the turning point of a quadratic using $x = -\frac{b}{2a}$. e.g. Determine the turning point coordinates of the parabola given by $y = 2x^2 + 8x - 5$.	☐
7E	12. I can sketch a quadratic graph using the quadratic formula. e.g. Sketch the graph of the quadratic $y = 3x^2 - 6x - 2$, labelling significant points. Round the x -intercepts to two decimal places.	☐
7F	13. I can apply quadratic models in word problems. e.g. A piece of wire measuring 80 cm in length is bent into the shape of a rectangle. Let x cm be the breadth of the rectangle. i Use the perimeter to find an expression for the length of the rectangle in terms of x . ii Hence, find a rule for the area of the rectangle, $A \text{ cm}^2$, in terms of x and sketch its graph for suitable values of x . iii Use the graph to determine the maximum area that can be formed and the dimensions of the rectangle that give this area.	☐
7G	14. I can find the points of intersection of a line and a parabola. e.g. Find the points of intersection of $y = 2x^2 - 5$ and $y = x + 1$.	☐
7G	15. I can determine the number of points of intersection of a line and a parabola. e.g. Determine the number of solutions (points of intersection) of the equations $4x + y = -4$ and $y = x^2 + 2x + 5$.	☐
7H	16. I can graph height vs time to show the rate of change of height. e.g. Water is poured into these containers at a constant rate. Draw a graph for each showing the relationship between the height, h , of water in the container at time t . i  ii 	☐

7H

17. I can work with distance-time graphs.

e.g. The distance-time graph shows a journey by scooter over a number of minutes.



- Between what points does the graph show a speed which is constant?
- Between what points does the graph show a speed which is decreasing?
- Between point C and E, at which point is the scooter traveling the fastest?

☐

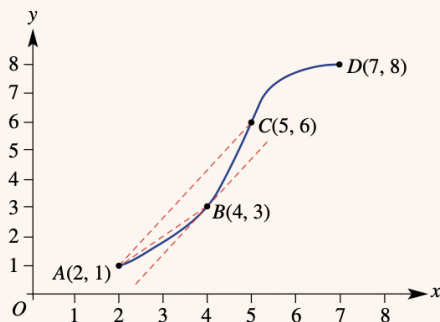
7I

18. I can find the average rate of change.e.g. Find the average rate of change of y in the rule $y = x^2 + 2$ as x changes from 1 to 3.☐

7J

19. I can approximate an instantaneous rate of change.

e.g. This graph shows part of a curve, with a tangent drawn at point B to illustrate the instantaneous rate of change at that point. Find the gradient of line segments AB and AC and state which one gives a better approximation of the instantaneous rate of change at point B.

☐

7J

20. I can work with direct variation.e.g. If y is directly proportional to x and $y = 45$ when $x = 25$, determine the relationship between y and x and use this to find y when $x = 40$.☐

7J

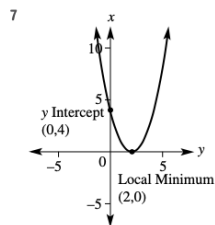
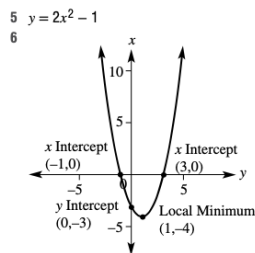
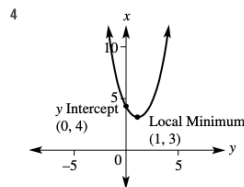
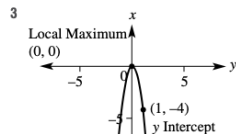
21. I can work with inverse variation.e.g. If x and y are inversely proportional and $y = 5$ when $x = 4$, determine the constant of proportionality k and write the rule.☐

Success criteria example questions

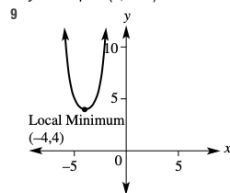
- 1 Turning point is a maximum at (2, 9)
Axis of symmetry is
x-intercepts are (-1, 0) and (5, 0)
y-intercept is (0, 5)

2

Formula	Max or min	Turning point	y-value when $x = 1$
$y = -x^2 + 2$	Max	(0, 2)	1

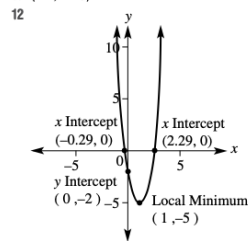


- 8 Turning point is at (1, -18)
x-intercepts are (-2, 0) and (4, 0)
y-intercept is (0, -16)



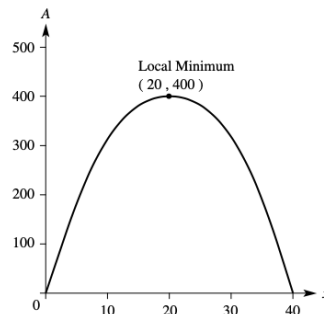
- 10 There are no x-intercepts

- 11 (-2, -13)



- 13 i $L = 40 - x$;

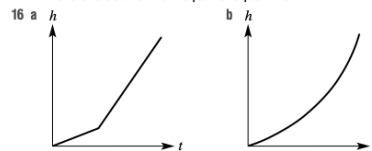
- ii $A = x(40 - x)$



- iii Maximum area is 400 cm² when $x = 20$, breadth = 20 cm and length = 20 cm.

- 14 $x = 2$, $y = 3$ and $x = -1.5$, $y = -0.5$

- 15 There is one solution to the pair of equations.



- 17 i A to B and B to C

- ii D to E

- iii A

- 18 average rate of change = 4

- 19 $m_{AB} = 1$; $m_{AC} = \frac{5}{3}$; AC gives a better approximation as the gradient of AC is closer to the instantaneous rate of change at B.

- 20 72

- 21 $k = 20$, $y = \frac{20}{x}$

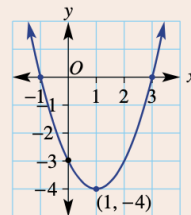
Short-answer questions

- 1 a Minimum at (1, -4) b $x = 1$
c (-1, 0) and (3, 0) d (0, -3)

Short-answer questions

7A

- 1 State the following features of the quadratic graph shown.
- turning point and whether it is a maximum or a minimum
 - axis of symmetry
 - coordinates of the x -intercepts
 - coordinates of the y -intercept



7A/B

- 2 State whether the graphs of the following quadratics have a maximum or a minimum turning point and give its coordinates.
- $y = (x - 2)^2$
 - $y = -x^2 + 5$
 - $y = -(x + 1)^2 - 2$
 - $y = 2(x - 3)^2 + 4$

7C

- 3 Sketch the quadratics below by first finding:

- the y -intercept
- the x -intercepts, using factorisation
- the turning point

a $y = x^2 - 4$ b $y = x^2 + 8x + 16$ c $y = x^2 - 2x - 8$

7D

- 4 Complete the following for each quadratic below.

- State the coordinates of the turning point and whether it is a maximum or a minimum.
- Find the coordinates of the y -intercept.
- Find the coordinates of the x -intercepts (if any).
- Sketch the graph, labelling the features above.

a $y = -(x - 1)^2 - 3$ b $y = 2(x + 3)^2 - 8$

7D

- 5 Sketch the following quadratics by completing the square. Label all key features with exact coordinates.

a $y = x^2 - 4x + 1$ b $y = x^2 + 2x + 6$ c $y = x^2 + 3x - 2$

7D/E

- 6 State the number of x -intercepts of the following quadratics either by using the discriminant or by inspection where applicable.

a $y = (x + 4)^2$ b $y = (x - 2)^2 + 5$
c $y = x^2 - 2x - 5$ d $y = 2x^2 + 3x + 4$

7E

- 7 For the following quadratics:

- find the coordinates of the y -intercept.
- use $x = -\frac{b}{2a}$ to find the coordinates of the turning point.
- use the quadratic formula to find the x coordinates of the x -intercepts, rounding to one decimal place.
- sketch the graph.

a $y = 2x^2 - 8x + 5$ b $y = -x^2 + 3x + 4$



7G

8 Solve these equations simultaneously.

a $y = x^2 + 4x - 2$
 $y = 10$

b $y = 2x^2 + 5x + 9$
 $y = -x + 4$

c $y = x^2 + 1$
 $2x + 3y = 4$

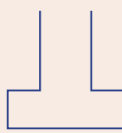
7G

9 Use the discriminant to show that the line $y = x + 4$ intersects the parabola $y = x^2 - x + 5$ in just one place.

7H

10 Water is being poured into these containers at a constant rate. Draw a graph showing the relationship between the height, h , of water in the container at time t .

a



b

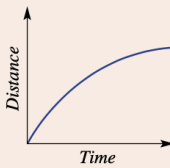


c



7H

11 For the distance–time graph showing a journey from home, describe the journey in relation to distance from home, gradient of graph and speed.



7I

12 For the relations with rules below, find the average rate of change of y as x changes from

i 0 to 2

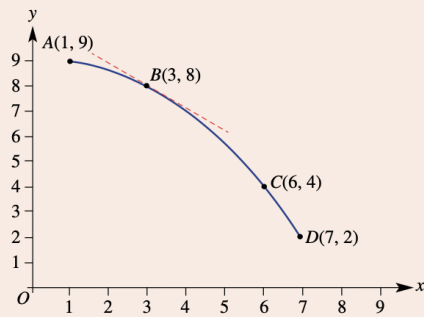
ii -1 to 4

a $y = x^2 + 2x$

b $y = -x^2 - 2x + 3$

7I

13 This graph shows part of a curve with a tangent drawn at point B to illustrate the instantaneous rate of change at the point. Use the points A to D to find the gradient of the line segment that gives the best approximation of the instantaneous rate of change at point B .



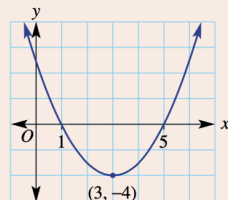
7J

14 a If y varies directly with x and $y = 10$ when $x = 2$, find the rule linking y with x .b If $y = \frac{k}{x}$ and $x = 6$ when $y = 12$, determine:i the value of k ii y when $x = 4$ iii x when $y = 0.7$

Multiple-choice questions

7A

1 The equation of the axis of symmetry of the graph shown is:

A $y = -4$ B $x = 3$ C $x = -4$ D $y = 3$ E $y = 3x$ 

7B

2 Compared to the graph of $y = x^2$, the graph of $y = (x - 3)^2$ is:

A translated 3 units down

B translated 3 units left

C dilated by a factor of 3

D translated 3 units right

E translated 3 units up

7B

3 The coordinates and type of turning point of $y = -(x + 2)^2 + 1$ is:A a minimum at $(-2, -1)$ B a maximum at $(2, 1)$ C a minimum at $(2, 1)$ D a maximum at $(2, -1)$ E a maximum at $(-2, 1)$

7B

4 The y -intercept of $y = 3(x - 1)^2 + 4$ has coordinates:A $(0, 1)$ B $(0, 4)$ C $(0, \frac{1}{3})$ D $(0, 7)$ E $(0, 3)$

7C

5 The x -intercept(s) of the graph of $y = x^2 + 3x - 10$ have x -coordinates at x equals:

A 2, -5

B -10

C 5, -2

D -5, -2

E 5, 2

7C

6 A quadratic graph has x -intercepts with x -coordinates at $x = -7$ and $x = 2$. The x -coordinate of the turning point is:A $x = -\frac{5}{2}$ B $x = -\frac{7}{2}$ C $x = \frac{9}{2}$ D $x = \frac{5}{2}$ E $x = -\frac{9}{2}$

7D

7 The quadratic rule $y = x^2 - 4x - 3$, when written in turning point form, is:A $y = (x - 2)^2 - 3$ B $y = (x - 4)^2 + 1$ C $y = (x - 2)^2 - 7$ D $y = (x + 4)^2 - 19$ E $y = (x + 2)^2 - 1$

7E

8 A quadratic graph $y = ax^2 + bx + c$ has two x -intercepts. This tells us that:

A The graph has a maximum turning point.

B $-\frac{b}{2a} < 0$ C There is no y -intercept.D $b^2 - 4ac > 0$ E $b^2 - 4ac = 0$

7F

- 9 A toy rocket follows the path given by $h = -t^2 + 4t + 6$, where h is the height above ground, in metres, t seconds after launch. The maximum height reached by the rocket is:

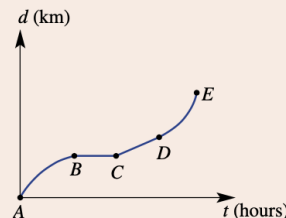
A 10 metres B 2 metres C 6 metres
D 8 metres E 9 metres

7H

- 10 The distance–time graph shows a journey by bicycle over a number of hours.

Between what points does the graph show a speed which is increasing?

A A to B
B C to D
C D to E
D C to D and D to E
E C to D and A to B



7I

- 11 The average rate of change of y as x changes from 1 to a ($a > 1$) in the rule $y = x^2 + 2$ is 4. The value of a is:

A 2 B 3 C 4 D 5 E 6

7J

- 12 If y is inversely proportional to x , the equation is of the form:

A $y = kx$ B $y = kx + c$ C $y = \frac{x}{k}$
D $y = kx^2$ E $y = \frac{k}{x}$

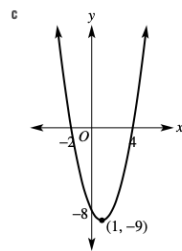
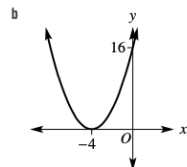
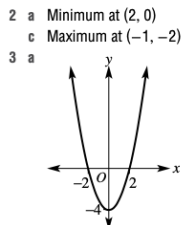
Extended-response questions

- 1 The cable for a suspension bridge is modelled by the equation $h = \frac{1}{800}(x - 200)^2 + 30$, where h metres is the distance above the base of the bridge and x metres is the distance from the left side of the bridge.
 - a Determine the turning point of the graph of the equation.
 - b If the bridge is symmetrical, determine the suitable values of x .
 - c Determine the range of values of h .
 - d Sketch a graph of the equation for the suitable values of x .
 - e What horizontal distance does the cable span?
 - f What is the closest distance of the cable from the base of the bridge?
 - g What is the greatest distance of the cable from the base of the bridge?
- 2 200 metres of fencing is to be used to form a rectangular paddock. Let x metres be the breadth of the paddock.
 - a Write an expression for the length of the paddock in terms of x .
 - b Write an equation for the area of the paddock ($A \text{ m}^2$) in terms of x .
 - c Decide on the suitable values of x .
 - d Sketch the graph of A versus x for suitable values of x .
 - e Use the graph to determine the maximum paddock area that can be formed.
 - f What will be the dimensions of the paddock to achieve its maximum area?

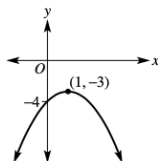
Short-answer questions

- 1 a Minimum at $(1, -4)$
c $(-1, 0)$ and $(3, 0)$

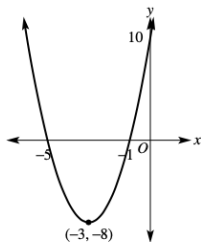
- b $x = 1$
d $(0, -3)$



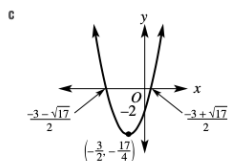
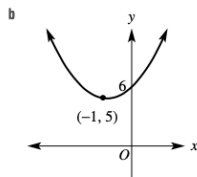
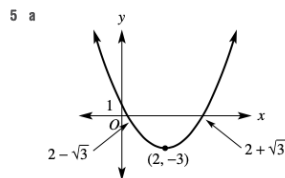
- 4 a i Maximum at $(1, -3)$
iii No x -intercepts
iv



- b i Minimum at $(-3, -8)$
iii $(-1, 0)$ and $(-5, 0)$
iv

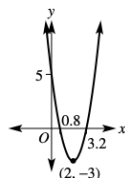


- b Maximum at $(0, 5)$
d Minimum at $(3, 4)$

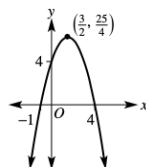


- 6 a 1
b 0
c 2
d 0

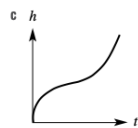
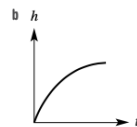
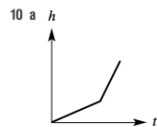
- 7 a i $(0, 5)$
ii $(2, -3)$
iii 0.8 and 3.2
iv



- b i $(0, 4)$
ii $(\frac{3}{2}, \frac{25}{4})$
iv



- 8 a $x = 2, y = 10$ and $x = -6, y = 10$
b No solutions
c $x = \frac{1}{3}, y = \frac{10}{9}$ and $x = -1, y = 2$
9 Show $b^2 - 4ac = 0$.



- 11 Increasing distance from home, decreasing positive gradient and decreasing speed.

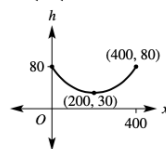
- 12 a i 4
b i -4
13 AC, -1
14 a $y = 5x$
b i $k = 72$
ii 5
ii -5
iii $\frac{720}{7}$

Multiple-choice questions

- 1 B 2 D 3 E 4 D 5 A 6 A
7 C 8 D 9 A 10 C 11 B 12 E

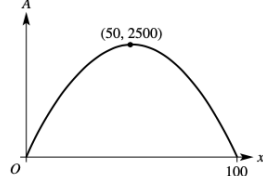
Extended-response questions

- 1 a $(200, 30)$
b $0 \leq x \leq 400$
c $30 \leq h \leq 80$
d



- e 400 m
f 30 m
g 80 m

- 2 a $100 - x$
b $A = x(100 - x)$
c $0 < x < 100$
d



- e 2500 m^2
f 50 m by 50 m